Simplify.

11) $\sin -6\theta \cos 4\theta + \cos -6\theta \sin 4\theta$

12) $\sin \theta \cos 3\theta + \cos \theta \sin 3\theta$

13) $\cos 3x \cos -4x + \sin 3x \sin -4x$

14) $\dfrac{\tan u + \tan 4u}{1 - \tan u \tan 4u}$

Write each trigonometric expression as an algebraic expression.

15) $\sin \left(\cos^{-1} \dfrac{\sqrt{2}}{2} - \sin^{-1} 2x \right)$

16) $\sin \left(\sin^{-1} 4x + \cos^{-1} x \right)$

17) $\cos \left(\tan^{-1} 2x + \sin^{-1} 0 \right)$

18) $\tan \left(\tan^{-1} 1 - \tan^{-1} \sqrt{x} \right)$

Verify each identity.

19) $\sin \left(\dfrac{\pi}{2} + \theta \right) = \cos \theta$

20) $\cos (\pi + \theta) = -\cos \theta$